MIL-I-60016A (MU)
7 November 1968
SUPERSEDING
MIL-I-60016 (MU)
31 January 1964

MILITARY SPECIFICATION

INCREMENTS, PROPELLANT FOR CHARGE, PROPELLING, M36A1 (MADE FROM PROPELLANT, M8)

1. SCOPE

1.1 This specification covers propellant increments made from M8 Propellant for use in charge, propelling M36A1 for 4.2 Inch Mortar Cannon M30.

1.2 Classification.-The propellant increments shall be of the following designations as specified (see 6.1).

Class 1 - Half increment

Class 2 - Increment

Class 3 - Five-Increment Bundle

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Flat; Steel

MILITARY

MIL-B-2427 - Box, Ammunition Packing; Wood, Nailed
MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-B-3106 - Board, Composition, Water-Resistant, Solid
MIL-L-10547 - Liners, Case, Waterproof, Flexible
MIL-B-13239 - Barrier Material, Waterproofed, Flexible All Temperature
MIL-I-45607 - Inspection Equipment, Supply and Maintenance

FSC: 1376

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:04Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-652 - Propellants, Solid, for Cannons; Requirements and Packing
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ORDNANCE CORPS

8796522 - Marking Diagram and Sealing for Wooden Packing Boxes
20-4-619 - Marking Diagram and Sealing of Fiberboard Cartons for Shipment of Propellant Increments
76-4-46 - Box, Packing, with Metal Liner, M24 for Smokeless Powder; Assembly and Details
76-4-56 - Box, Packing, Metal-Wood, M17 for Smokeless Powder
76-4-59 - Box, Packing, Metal-Wood, M18 for Smokeless Powder Assembly
7548645 - Carton, Packing, Reusable Collapsible for High Explosive, Assembly Details, Packing and Marking
7549033 - Container, Metal, Universal, M25 for Propellants and Explosives
8858577 - Marking Diagram and Sealing of Container, Metal, Universal M25 for Shipment of Propellants
8858848 - Marking Diagram and Sealing of Metal Lined Wooden Packing Boxes for Shipment of Propellants
8833726 - Half-Increment
8833728 - Increment
8833727 - Five-Increment Bundle
8863617 - Charge, Propelling; M36A1 Assembly

PUBLICATIONS

ORDNANCE CORPS

8863617 - Index of Inspection Equipment Lists

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:04Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS, Title 49 - Transportation, Parts 171-179

The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations, available from the Superintendent of Documents, Government Printing Office, Washington, D. C., 20402. Orders for the above publication should cite: 49 CFR 171-179 (latest revision).

3. REQUIREMENTS

3.1 Material.-The propellant shall be M8 and shall conform to the requirements of MIL-STD-652 and the applicable drawings.

3.2 Individual increment weights.-The increments shall be adjusted to the proper weight of the charge established by the ballistic test within the tolerances specified in Table I.

TABLE I

Increments	Individual Increment Weights for M36A1 Propelling Charge
Half-increment	1/72 charge: ± 2 grains
Increment	2/72 charge: ± 3 grains
Five increment	10/72 charge: ± 12 grains

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:04Z
Check the source to verify that this is the current version before use.

3.3 Ballistics.-The propellant, when fired in the weapon and with the weight of projectile for which it was manufactured, shall comply with the following requirements when tested as specified in 4.3.

3.3.1 Velocity and pressure.-At 70 degrees Fahrenheit, a weight of propellant assessed as required to impart a muzzle velocity as specified in Table II to the projectile shall not exceed a pressure specified in Table II at the 41 increment level.

TABLE II

With Container Extension at 26.23 ± 0.01 pounds

No. of Increments	Muzzle Velocity feet per second (fps) Min.	Max.	Upper Pressure Limit, pounds per square inch (psi)
41	955	1002	15600

3.3.2 Uniformity of velocity with a plastic walled M2A2 Ignition Cartridge and with Container Extension, projectile weight of 26.23 pounds.-With a charge of forty one (41) increments, the standard deviation of velocity for the lot at 70 degrees Fahrenheit shall not exceed 6.5 feet per second (see 4.2.3.1.2.2).

3.3.3 Permissible individual maximum pressure.-No individual maximum pressure shall be greater than 17,900 psi (115 percent of the upper pressure limit of the weapon) when fired at 70 degrees Fahrenheit (see 4.2.3.1.2.4).

3.4 Workmanship.-The best commercial practices shall be used in the formulation of the propellant furnished under this specification and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. The propellant increments shall show no visible evidence of moisture and shall be free of cracked, torn, deformed or otherwise distorted propellant sheets. The stitches, bags and seals shall be without breaks, tears, grease or noticeable foreign matter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:04Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Number of batches of product inspected (shall be made available upon request by the contracting officer).

c. Results obtained, by defect code, for all inspections performed (shall be made available upon request by the contracting officer).

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents (shall be made available upon request by the contracting officer).

f. Number of items in the lot.

g. Date submitted.

h. Eight copies of propellant description sheet AMC Form 1047 (00 Form 1204) containing the applicable data (see 6.6).

4.1.2 The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:04Z
Check the source to verify that this is the current version before use.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "lot" as used throughout this specification, refers to an inspection lot, which is defined as an essentially homogeneous quantity of propellant from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. A lot shall consist of one or more batches of M8 propellant produced by one manufacturer, in accordance with the same specification, and same specification revision under one continuous set of operating conditions. Each lot shall consist of that quantity of M8 propellant which has been subjected to the same unit chemical or physical process intended to make the final product homogeneous.

4.2.2 Examination.-Inspection for critical defects (and major defects, when so specified), shall be 100 percent. Sampling plans and procedures for major and minor defects shall be in accordance with Standard MIL-STD-105 except that continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each minor defect and an AQL of 0.40 for each major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with paragraph 4.2.4.

4.2.2.1 Assembly, increment or 1/2 increment (see drawing 8833727, 8833728, or 8833726).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.65 percent		
101.	Weight outside tolerance	Approved Scale	01001
102.	Hole missing	Visual	01002
103.	Diameter of holes other than prescribed	Gage	01003
104.	Length of sides other than prescribed	Gage	01004
106.	Evidence of poor workmanship (see 3.4)	Visual	01006
107.	Incorrect number of increments or half-increments	Visual Count	01007

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:04Z
Check the source to verify that this is the current version before use.

Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 4.00 percent		
201.	Foreign matter on propellant	Visual	01008
202.	Insufficient number of stitches per inch	Visual	01009
203.	Sheet bent over	Visual	01010
204.	Stitches too close to edge (possibility of tearing during handling)	Visual	01011
205.	Radial cuts not in same plane	Visual	01012
207.	Insufficient number of stitches per increment	Visual	01014

4.2.2.2 Sealed inner liner (see drawing 7548645).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Liner damaged so that contents are exposed or liable to become exposed	Visual	02001
102.	Bag incompletely sealed	Visual	02002
Minor:	None defined.		

4.2.2.3 Sealed wood packing box (see paragraph 5.1.1 if applicable). Sealed fiber board carton (see drawing 20-4-619 if applicable).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Banding or strapping missing, broken or loose	Visual-Manual	03001
102.	Board broken or split	Visual	03002
103.	Contents exposed	Visual	03003

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:04Z
Check the source to verify that this is the current version before use.

Minor:	AQL 1.50 percent		
201.	Banding or strapping improperly engaged	Visual-Manual	03004
202.	Contents move when shaken	Manual	03005
203.	Marking missing, misleading, or illegible	Visual	03006
204.	Nails, screws or fasteners loose	Visual	03007

4.2.2.4 Box or container, prior to filling (see drawing 76-4-46, 76-4-59, 7549033, or 76-4-56, as applicable).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Cracks or holes in metal liner (as applicable)	Visual	04001
102.	Foreign matter on interior	Visual	04002
Minor:	None defined.		

4.2.2.5 Sealed box or container (see drawing 76-4-46, 76-4-59, or 76-4-56, 7549033, 8858848 or 8858577, as applicable).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Cover incompletely closed	Visual-Manual	05001
102.	Gasket missing or broken	Visual	05002
103.	Wooden cover split or otherwise damaged (when applicable)	Visual	05003
Minor:	AQL 2.50 percent		
201.	Car seal missing, unsealed or improperly positioned	Visual-Manual	05004
202.	Crack or hole in metal container	Visual	05005
203.	Hardware improperly engaged	Visual	05006
204.	Marking misleading or unidentifiable	Visual	05007
205.	Bare areas in exterior coating of metal container the sum of which is in excess of 1/4 square inch	Visual	05008

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:04Z
Check the source to verify that this is the current version before use.

4.2.2.6 Final Pull

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	100% Inspection		
101.	Thickness other than prescribed (see 4.2.3.3).....	Gage	07001

4.2.3 Testing

4.2.3.1 Sampling.-Ten containers shall be randomly selected from the lot. Where the lot consists of fewer than eleven sublots, ten containers shall comprise approximately equal numbers from each sublot. Where the number of sublots is eleven or more, one container shall be selected from each of ten randomly chosen sublots. Sublot identities shall be marked on each container.

4.2.3.1.1 For composition and surveillance, the sampling shall be in accordance with Standard MIL-STD-652.

4.2.3.1.2 Ballistic sample.-An approximately equal number of increments shall be withdrawn from each of the ten containers to provide the minimum ballistic sample indicated in Table III. Each portion of the sample shall be packed in a separate air tight container and labeled in accordance with Standard MIL-STD-652. A ten round uniformity series shall be fired after conditioning the rounds at 70 degrees Fahrenheit for each number of increments specified in Table III.

TABLE III

	M8 Ballistic Sample No. Increment, Minimum		
No. of Increments	1/2 Increment	1 Increment	5 Increment Bundle
7	--	20	--
41	20	50	60

To allow for retest, the above ballistic sample must be doubled.

4.2.3.1.2.1 Velocity (critical defect).-The sampling plan for this test shall be in accordance with Standard MIL-STD-414, Table B3, Code Letter E and an AQL of 0.65 percent (see 3.3.1), Code No. 06001.

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:04Z
Check the source to verify that this is the currentversion before use.

4.2.3.1.2.2 Velocity uniformity (Major defect).-The lot shall be rejected if the standard deviation of the uniformity series times the factor 0.73 fails to comply with the applicable requirements (see 3.3.2), Code number 08001.

4.2.3.1.2.3 Upper pressure limit, Critical defect.-The lot of propellant shall be rejected if the maximum pressure corresponding to the recommended charge weight exceeds the applicable requirement, (see 3.3.1), Code No. 09001.

4.2.3.1.2.4 Permissible individual maximum pressure, Critical defect.-If the mean of the maximum pressures obtained during the uniformity series firing at 70°F plus three (3) times the standard deviation exceeds the applicable requirement, the lot shall be rejected (see 3.3.3), Code No. 10001.

4.2.3.2 Retest.-If for any reason the proof director considers that test conditions have detrimentally affected the test results, he shall declare the results invalid and perform a new test on additional charges as required (see 6.10).

4.2.3.3 Final Roll Sheets.-Check every pull from each final roll by measuring four (4) randomly selected sheets and measure the sheet in four (4) places (right side, left side, tail end and pull end). Measurements should be approximately at the center of the edge being measured. If any measurement is outside the limit, the pull shall be rejected.

4.2.4 Inspection equipment.-The contractor shall have available and utilize correctly all necessary standard, Government approved measuring equipment required for the performance of the inspections specified herein.

4.3 Test methods and procedures

4.3.1 Acceptance tests.-These tests shall be conducted at a Government Proving Ground in accordance with the applicable U. S. Army Test and Evaluation Command Materiel Test Procedures, TECP-700-700, Interim Pamphlet.

4.3.2 Ballistics.-Ten (10) rounds shall be loaded with sufficient increments (see 4.2.3.1.2) randomly chosen from the test lot and then conditioned to constant temperature at 70 degrees F. Ten (10) standard reference rounds shall be similarly loaded, but with reference propellant, and conditioned. No more than five minutes shall elapse from the time the round is removed from the conditioning chamber until it is fired. The charges shall be prepared in a manner specified in drawing number 8863617 with M329A1 projectiles, plastic walled M2A2 ignition cartridges, and M9 propellant bags. This uniformity series shall be fired at the forty-one (41) increment level and the temperature of the day at time of firing recorded.

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:04Z
Check the source to verify that this is the current version before use.

The velocity mean and standard deviation shall be calculated after reducing results to standard conditions and all firing data, including individual pressure readings, shall be reported to Picatinny Arsenal for evaluation (see 6.3). In addition, a series of ten (10) rounds shall be fired at the seven (7) increment level for information only. Both reference and test rounds at both increment levels shall be loaded with the same lot of standard M9 propellant.

5. PREPARATION FOR DELIVERY

5.1 Packing (see 6.1).

5.1.1 Level A.-Propellant increments shall be packed in wooden packing boxes that conform to Grade A, Type 1, Class 1, Specification MIL-B-2427, except that bottom, sides and top may be 1/2 inch thick lumber. The inside dimensions of the box shall be 14 1/2 plus 1/8 inches long by 14 1/2 plus 1/8 inches wide by 7 1/4 plus 1/8 inches deep. Each wooden packing box shall be fitted with a caseliner conforming to Type I, Grade B, Style A, Specification MIL-L-10547; manufactured from barrier material conforming to Grade A, Type CW-2, Class I, Specification MIL-B-13239. The caseliner shall be approximately 14 3/4 inches long by 14 3/4 inches wide by 18 1/2 inches deep. Both ends and top of the wooden box shall be lined between the box and the caseliner, with fillers manufactured of fiberboard conforming to Type I or II, Specification MIL-B-3106. The cover shall be secured to the box with nails, or wood screws in accordance with Specification MIL-B-2427. The box shall be strapped girthwise, between cover cleats with two 5/8 inch by .02 inch thick flat steel straps conforming to Type I, Class A or B, Specification QQ-S-781. The net weight shall not exceed 50 pounds of propellant.

5.1.2 Level B.-Propellant increments shall be packed in propellant packing boxes M17, M24, or M25, complying with drawing 76-4-46, 76-4-59, 76-4-56, or 7549033, respectively. Containers shall be closed, sealed and tested in accordance with drawing 8858848.

5.1.3 Level C.-Propellant increments shall be packed in accordance with drawing 7548645, Carton No. 1, except that the bag liner shall be manufactured from barrier material conforming to Grade A, Type CW-2, Specification MIL-B-13239.

5.2 Marking

5.2.1 Level A.-Exterior containers shall be marked in accordance with drawing 8796522. The ICC nomenclature shall be "Propellant Explosive - Class A."

Source: https://assist.dla.mil -- Downloaded: 202640-23T00:04Z
Check the source to verify that this is the currentversion before use.

5.2.2 Level B.-The M17, M18, and M24 containers shall be marked in accordance with drawing 8858848. The M25 container shall be marked in accordance with drawing 8858577.

5.2.3 Level C.-The fiberboard carton shall be marked in accordance with drawing 20-4-619.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Packing required (see 5.1).
c. Process.-Details of the manufacturing process and the equipment used by the contractor will be submitted to the procuring activity in writing prior to commencement of manufacture.

6.2 Mean and standard deviation.-The mean and standard deviation shall be calculated in a standard statistical technique equivalent to that shown in Standard MIL-STD-414, Section B, Example B-1. (n-1) shall be used as a divisor in calculating the standard deviation.

6.3 Submission of test data.-In addition to the normal distribution of records, when the propellant is procured by the Department of the Army, one copy of all ballistic test data shall be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND2, and one copy to ATTN: SMUPA-DR2, Dover, New Jersey.

6.4 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.5 Factor 0.73.-The factor 0.73 makes allowance for the probability that a standard deviation from a ten-round sample exceeds the true standard deviation by chance alone.

6.6 Submission of AMC Form 1047.-Information contained in propellant description sheet, AMC Form 1047, need not be resubmitted in the form of a certificate provided that the provisions regarding the certifying of responsible agents are adhered to (see 4.1.2).

6.7 Standard deviation.-The standard deviation used is the calculated value obtained for the particular lot tested.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:04Z
Check the source to verify that this is the current version before use.

6.8 Tolerance.-A tolerance of plus or minus 2 degrees F applies to all 70 degree F requirements.

6.9 Upper pressure limit for propellant is that value of the pressure which is specified in the propellant specification as the upper limit of average pressure at 70 degrees F, which may be developed by an acceptable propellant, in the form of propelling charges, which will impart the specified muzzle velocity to the specified projectile at its standard zone weight, under stated cannon conditions.

6.10 If the mean of the maximum pressures obtained during the uniformity series complies with the applicable requirement (see 4.2.3.1.2.4) but one or more individual rounds exceeds 17,900 psi, the lot shall be referred to Picatinny Arsenal for disposition.

6.11 Surveillance samples.-Samples shall be forwarded to Picatinny Arsenal, Dover, New Jersey, ATTN: SMUPA-VG, for the propellant surveillance test program in accordance with MIL-STD-652.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1376-A-023

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:04Z
Check the source to verify that this is the current version before use.